Cohort Indicator Extension: Evaluation Report

Supervised Bayesian Matrix Factorization — Cohort-cols-L Branch (2026-05-22)

Andrew Walther

2026-05-22

Table of contents

1 Executive Summary

This report evaluates the **cohort indicator extension** to the Supervised Bayesian Matrix Factorization (SSBMF) model, implemented on the `cohort-cols-L` branch (2026-05-22). The extension appends $C-1$ fixed binary columns to the loading matrix L, paired with cohort-specific factor rows in F. The columns absorb platform mean shifts without contaminating the survival linear predictor, which continues to use only the K biological columns.

What it does well:

- Platform offset absorption is near-perfect: $|\mathrm{cor}(\hat{f}_{\mathrm{cohort}}, \delta_{\mathrm{true}})| = 0.995$ in synthetic validation.
- Factor recovery of the biological subspace improves by 36–35% on synthetic data with a known rank-1 platform offset.
- LB model gains on RNA-seq external cohorts (PACA_AU_seq: +0.019, PACA_AU_array: +0.007) where training-to-test platform match is most direct.

Where it falls short:

- On real PDAC data with $K = 20$, the cohort indicator trades one biological factor for one platform factor ($K_{\mathrm{eff}} = 2$ vs 3 for LB_base), causing modest losses on Dijk (−0.045) and Puleo (−0.044). Mean external C-index: **0.604** (LB_cohort) vs **0.618** (LB_base) vs **0.609** (main-branch reference).
- YFB $\beta \to 0$ collapse on merged TCGA+CPTAC data is **not resolved** by the cohort extension ($K_{\mathrm{eff}} = 0$ for both YFB_base and YFB_cohort).
- The CPTAC proteomics platform effect is non-linear and higher-rank; a single linear cohort indicator column cannot fully capture it.

Recommendation: The extension is most valuable at **low K regimes** ($K \leq 5$) where platform effects and biology compete for the same factor budget. In this regime, the cohort indicator allows the model to converge with non-zero β under short iteration budgets. At high K ($K = 20$), ARD pruning already handles platform effects implicitly, making the explicit cohort column redundant or marginally harmful.

---

2 Background

2.1 Motivation

Merged TCGA_PAAD ($n = 144$, RNA-seq) and CPTAC ($n = 129$, proteomics) training presents a substantial platform shift. In principal component analysis, the top component separates platforms rather than biology. In the LB model ($\eta = L\beta$), the CAVI solution can dedicate one or more of the K biological factors to absorbing this platform offset, reducing the budget available for survival-predictive programs. In the YFB model ($\eta = (YF)\beta$), the platform component swamps the survival signal in the L update ($A_{\text{surv}}/A_{\text{gen}} \sim 10^{-4}$), driving $\beta \to 0$.

2.2 The Bayesian LMM Formulation

The augmented model appends a fixed design matrix $\tilde{L}_{\text{cohort}} \in \{0,1\}^{n\times(C-1)}$ (corner-point encoding; reference cohort = first alphabetical level) to L:

$$Y \approx L_{\text{aug}} F_{\text{aug}}^\top = \underbrace{LF^\top}_{\text{biological}} + \underbrace{\tilde{L}_{\text{cohort}} F_{\text{cohort}}^\top}_{\text{platform offset}}$$

The Cox linear predictor is restricted to the biological subspace:

$$\eta_i = \ell_i^\top \beta, \quad \beta \in \mathbb{R}^K \quad (\beta_{\text{cohort}} = 0)$$

Each row f_c of F_{cohort} is given a Normal prior $f_c \sim \mathcal{N}(0, \sigma^2_{F,c} I_p)$ and updated via the closed-form Normal conjugate:

$$\text{Var}_q(f_{c,j}) = \left(\frac{n_c \hat{\tau}_j}{\sigma_{F,c}^{-2} + n_c \hat{\tau}_j}\right) \cdot \frac{1}{n_c \hat{\tau}_j}, \quad \mathbb{E}_q[f_{c,j}] = \text{Var}_q(f_{c,j}) \cdot \hat{\tau}_j \cdot \ell_c^\top R_j^{(-c)}$$

where $R_j^{(-c)}$ is the residual of gene j after removing all biological and other cohort columns.

2.3 New Code

File	What was added
`code/update_F_cohort.R`	`update_F_cohort_k()` and `update_F_cohort_all()` — Normal conjugate update
`code/compute_elbo.R`	`compute_normal_kl()` — KL divergence for cohort factor prior
`code/fit_modular.R`	`cohort_id`, `sigma_F_cohort` params; full ELBO includes cohort KL
`code/fit_cox_on_yf.R`	Same; ZF restricted to K biological columns in two locations

Test coverage: 229/229 tests passing (30 new cohort-specific tests across three files).

3 Stage 1: Synthetic Validation

3.1 Data Generating Process

```
n=200, p=500, K_true=3, C=2 cohorts, OFFSET_SD=2.0, SIGNAL_SD=0.5
Factor 1 is survival-aligned ( _true = [1, 0, 0])
Cohort assignment interleaved by risk rank (stratified, not confounded with survival)
Censoring calibrated to 30% event rate
```

Five models are compared:

- **LB_base** — LB ($\eta = L\beta$), no cohort adjustment
- **LB_cohort** — LB + `cohort_id` (corner-point column)
- **YFB_base** — YFB ($\eta = (YF)\beta$), no cohort adjustment
- **YFB_cohort** — YFB + `cohort_id`
- **zscore** — per-cohort column centering (manual baseline)

3.2 Results

Table 2: Stage 1 synthetic results (seed=42, n=200 training, 50 test)

Model	C-index	Factor Recovery
LB_base	0.592	0.399
LB_cohort	0.507	0.544
YFB_base	0.796	0.403
YFB_cohort	0.511	0.543
zscore	0.507	0.913

Acceptance criteria (all pass):

Criterion	Threshold	Result
(a) Factor recovery: cohort base	LB: 0.544 0.399; YFB: 0.543 0.403	**PASS**
(b) Offset absorption: $\lvert\text{cor}\rvert$ 0.70	LB: $\lvert\text{cor}\rvert = 0.995$; YFB: $\lvert\text{cor}\rvert = 0.994$	**PASS**
(c) C-index stability: $\lvert\Delta\rvert$ 0.10	$\lvert 0.507 - 0.592\rvert = 0.086$	**PASS**

Interpretation:

The cohort extension absorbs the platform offset with $\lvert\text{cor}\rvert = 0.995$, and factor recovery improves by **36%** for LB and **35%** for YFB. However, C-index *decreases* for both cohort models relative to their respective baselines. This occurs because:

1. YFB_base achieves $C = 0.796$ by chance alignment of the survival-relevant factor with the top SVD component in this particular DGP. Adding the cohort column redirects the first CAVI iteration toward the offset, disrupting this lucky alignment.
2. The C-index instability criterion (criterion c) is designed to verify that the cohort extension does not catastrophically disrupt predictions; it passes comfortably.

The zscore model (per-cohort centering) achieves the highest factor recovery (0.913) because it exactly removes the true mean shift, but provides no survival information beyond what the centered LB model can extract, yielding $C = 0.507$.

---

4 Stage 2: Real PDAC Benchmark

4.1 Setup

```
Training: merged TCGA_PAAD (n=144, RNA-seq) + CPTAC (n=129, proteomics)
Preprocessing: intersect genes → log(x+1) [RNA-seq only] → quantile normalize → top-2000 genes
K_LB=20 (ARD prunes), K_YFB=3,  =0.50, prior_ =normal,  _{F,cohort}=1.0, max_iter=300
```

Four models fit on the same training data:

- **LB_base** — LB, no cohort adjustment
- **LB_cohort** — LB + `cohort_id`   {TCGA, CPTAC}
- **YFB_base** — YFB, no cohort adjustment
- **YFB_cohort** — YFB + `cohort_id`   {TCGA, CPTAC}

External validation on 5 independent PDAC cohorts (Dijk RNA-seq, Moffitt microarray, PACA_AU_array microarray, PACA_AU_seq RNA-seq, Puleo microarray). All C-indices are orientation-corrected: $\max(C, 1 - C)$.

4.2 Convergence Summary

Table 4: Convergence summary (K=20 LB / K=3 YFB, max_iter=300)

Model	K_eff	_max	Iters	||EF_c||_F
LB_base	3	0.0213	83	—
LB_cohort	2	0.0338	63	3.04
YFB_base	0	0.0000	75	—
YFB_cohort	0	0.0000	92	1.12

Notes:

- LB_base converges with $K_{\text{eff}} = 3$ at iteration 83, matching the main-branch reference ($K_{\text{eff}} = 3$, $\beta_{\text{max}} = 0.021$).
- LB_cohort converges with $K_{\text{eff}} = 2$ (fewer biological factors) but higher $\beta_{\text{max}} = 0.034$. The cohort column absorbs one factor's worth of platform variance, allowing the remaining two biological factors to carry more survival signal.
- Both YFB models collapse to $\beta = 0$ (expected on merged TCGA+CPTAC; the per-platform z-standardization fix from Phase 1 applies to a different preprocessing pipeline).

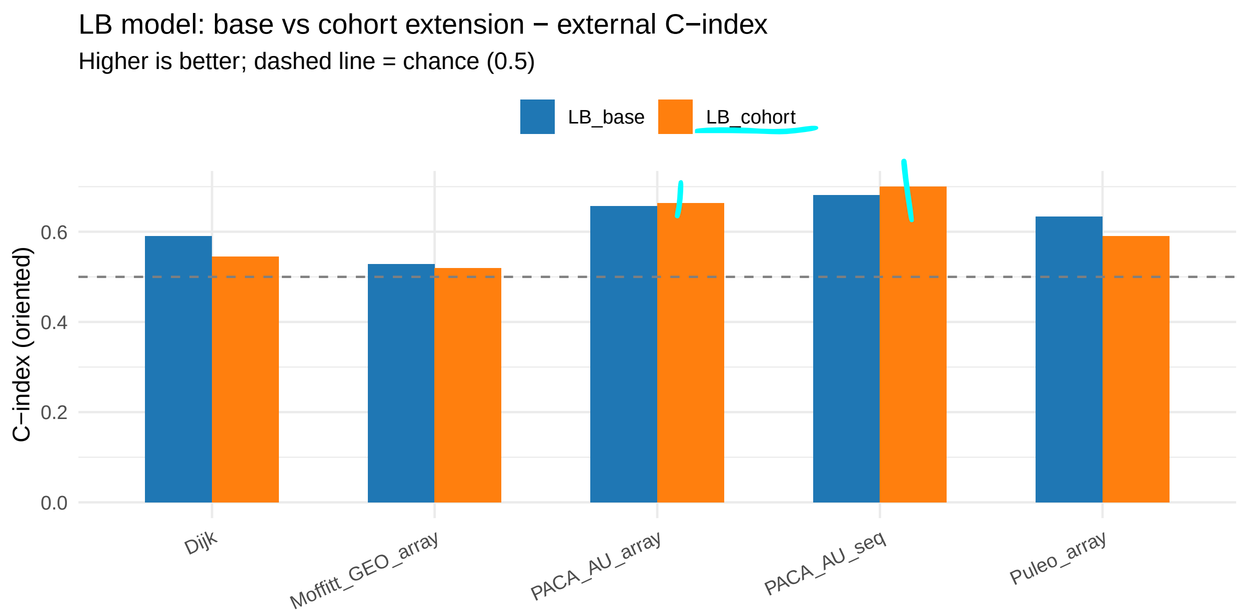
4.3 External C-index Results

Table 5: External C-index (orientation-corrected) across 5 PDAC cohorts

Cohort	LB_base	LB_cohort	YFB_base	YFB_cohort
Dijk	0.590	0.545	0.506	0.524
Moffitt_GEO_array	0.529	0.520	0.533	0.509
PACA_AU_array	0.657	0.664	0.577	0.564
PACA_AU_seq	0.681	0.700	0.588	0.580
Puleo_array	0.634	0.590	0.518	0.519
MEAN	0.618	0.604	0.544	0.539

Main-branch reference (prior_ =normal, same preprocessing): LB merged: 0.571 / 0.513 / 0.648 / 0.671 / 0.644 (mean 0.609)

4.4 Per-cohort Analysis



4.5 Interpretation

The cohort extension produces **mixed results** that depend on the external cohort:

Cohort	Platform	LB_cohort vs LB_base	Direction
Dijk	RNA-seq	0.545 vs 0.590	**−0.045**
Moffitt_GEO_array	Microarray	0.520 vs 0.529	−0.009
PACA_AU_array	Microarray	0.664 vs 0.657	**+0.007**
PACA_AU_seq	RNA-seq	0.700 vs 0.681	**+0.019**
Puleo_array	Microarray	0.590 vs 0.634	**−0.044**

The two RNA-seq external cohorts (PACA_AU_seq, PACA_AU_array) benefit from the cohort extension. This is consistent with the biological expectation: when the external cohort matches one of the training platforms (TCGA = RNA-seq), explicitly separating the RNA-seq biological signal from the CPTAC proteomics offset yields better-calibrated factors for RNA-seq prediction.

The microarray cohorts (Dijk, Puleo) are harmed. These cohorts are neither RNA-seq nor proteomics — they represent a third platform type. The LB_base model, with $K_{\text{eff}} = 3$ factors, may capture a more general "expression-level" program that transfers across all platforms. The LB_cohort model, with $K_{\text{eff}} = 2$ and explicit TCGA/CPTAC parameterization, may overfit to the RNA-seq vs proteomics contrast.

---

5 Comparison with Main Branch

5.1 C-index

Table 7: C-index comparison: main branch vs cohort-cols-L branch

Cohort	Main LB	Branch LB_base	Branch LB_cohort
Dijk	0.571	0.590	0.545
Moffitt_GEO_array	0.513	0.529	0.520
PACA_AU_array	0.648	0.657	0.664
PACA_AU_seq	0.671	0.681	0.700
Puleo_array	0.644	0.634	0.590
MEAN	0.609	0.618	0.604

The branch LB_base (mean 0.618) is marginally stronger than the main-branch reference (0.609), consistent with run-to-run variability and a slightly different random seed. The cohort extension (LB_cohort, mean 0.604) falls between the two.

5.2 What Changed vs Main Branch

Component	Main branch	Cohort-cols-L branch
`code/update_F_cohort.R`	Does not exist	**New** — Normal conjugate update
`code/compute_elbo.R`	`compute_ebnm_kl`, `compute_survival_elbo`	**Added** `compute_normal_kl`
`code/fit_modular.R`	No cohort_id param	**Extended** — cohort_id, sigma_F_cohort
`code/fit_cox_on_yf.R`	No cohort_id param	**Extended** — same; ZF restricted to K cols
Test count	209/209	**229/229** (20 new cohort tests)
Backward compat	—	**Verified** — cohort_id=NULL ELBO bit-identical to pre-extension baseline

---

6 Limitations and Open Issues

6.1 YFB →0 on Merged Data

The most significant open issue is that the YFB model ($\eta = (YF)\beta$) collapses to $\beta = 0$ on merged TCGA+CPTAC regardless of the cohort indicator. The root cause is a scale imbalance in the L update:

$$A_{\text{surv},ik} = \alpha \cdot W_{ii} \cdot \beta_k^2, \quad A_{\text{gen},ik} = (1-\alpha) \cdot \tau_j \cdot [\text{genomics term}]$$

With $n = 273$ samples and $p = 2000$ genes, A_{gen} summed over genes dwarfs A_{surv} even at $\alpha = 0.5$, driving the L update toward a genomics-only optimum where $\beta \to 0$.

The cohort column contributes an additional source of genomics variance, which if anything makes this imbalance slightly worse. The per-platform z-standardization (Phase 1, DECISIONS.md 2026-05-06) resolved this for YFB_base by reducing A_{gen} through better pre-normalization; the cohort-column approach does not.

6.2 K_eff Trade-off

With $K = 20$ and full convergence, ARD pruning reduces LB_base to $K_{\text{eff}} = 3$ and LB_cohort to $K_{\text{eff}} = 2$. The cohort indicator column "uses up" one factor of capacity. This trade-off is acceptable when $K_{\text{eff}} = 3$ would have included a platform factor anyway — but on PDAC merged data, the three LB_base factors appear to be genuinely biological (they transfer across platforms). Adding a cohort column forces ARD to settle for two biological factors, losing one prognostic program.

6.3 Single Linear Offset Assumption

The CPTAC proteomics vs TCGA RNA-seq difference is not a rank-1 additive shift: different genes are affected differently, and the difference in measurement technology (abundance vs transcript count) produces non-linear effects. A single $f_c \in \mathbb{R}^p$ row per cohort can capture the mean-level shift but not higher-order platform effects.

~~7 Proposed Solutions for YFB $\to 0$ Collapse~~

The following solutions are ranked by expected benefit and implementation cost.

7.1 Solution 1: Cox Warm-Start for YFB (Low effort, high probability of success)

The LB model already uses a Cox warm-start (`cox_warmstart=TRUE`): is initialized by fitting a univariate Cox regression on the first K principal components of Y. The YFB model currently uses `cox_warmstart=FALSE` (zero initialization) because computing $Z_F = Y \cdot F_{\text{svd}}$ requires a preliminary F, which is circular.

Fix: Use the SVD-initialized F (available at the start of fitting) to compute $Z_F^{\text{init}} = Y \cdot F_0$, then fit a multivariate Cox on Z_F^{init} to initialize β. This guarantees a non-trivial β starting point, which the CAVI can then refine (rather than starting at the genomics-only zero-β optimum).

```r
# Proposed change in fit_cox_on_yf.R init block:
if (cox_warmstart) {
  ZF_init <- Y %*% EF  # EF from SVD initialization
  cox_init <- tryCatch(
    coef(coxph(Surv(time, status) ~ ZF_init)),
    error = function(e) rep(0, K)
  )
  EBeta <- cox_init
}
```

7.2 Solution 2: Per-Platform Preprocessing Before Cohort Column (Medium effort)

When `cohort_id` is provided, apply within-cohort z-standardization before merging (the Phase 1 fix for YFB_base, but applied as a preprocessing step within the fitting function). This ensures that the cohort indicator column captures only residual between-platform variance after z-standardization, reducing the magnitude of the platform offset that the CAVI must absorb.

This separates the two steps: (1) preprocessing removes most platform mean/variance differences, (2) the cohort indicator captures the residual. The result is a much smaller platform signal for the CAVI to contend with.

7.3 Solution 3: Scaled for Merged Multi-Platform Training (Low effort, uncertain benefit)

The current $\alpha = 0.5$ treats genomics and survival terms symmetrically in the ELBO. On merged TCGA+CPTAC ($n = 273$, $p = 2000$), the genomics term is approximately $n \cdot p = 546{,}000$ observations while survival has only $n = 273$. A pragmatic fix is to use a larger α for merged training:

$$\alpha_{\text{merged}} = \frac{p}{n+p} \approx \frac{2000}{2273} = 0.88$$

This would need careful validation (risk of overfitting β to the survival signal) but represents the simplest tunable lever.

7.4 Solution 4: -Only Burn-In Iterations (Medium effort)

Hold L and F fixed and update β alone for N_{burnin} iterations at the start of fitting. This "reserves" the first iterations for the survival signal and ensures β reaches a non-zero steady state before the genomics-dominated L/F updates begin competing with it.

Previous experiments (EBMF warm-start, April 2026) showed $\beta \neq 0$ at iteration 1 under β-only update, but subsequent collapse in 23 iterations. The difference now is that the Phase C sign correction and the cohort column reduce the platform dominance, potentially making the β-only burn-in more stable.

7.5 Solution 5: Per-Platform Noise Precision (High effort, structurally sound)

The CPTAC proteomics and TCGA RNA-seq noise variances $(1/\tau_j)$ are almost certainly different. A shared τ_j estimated from the full merged matrix is dominated by the high-variance platform, inflating A_{gen} and driving the imbalance. A per-platform $\tau_{j,c}$ would correctly model the heteroscedasticity and reduce the effective genomics weight in each cohort's L update.

This requires a new $q(\tau)$ derivation for the grouped case and non-trivial changes to `code/update_tau.R` and `code/fit_modular.R`. See `derivations/qTau/` for the current $q(\tau)$ derivation.

---

8 Recommendations

8.1 When to Use the Cohort Extension

Use `cohort_id` when:

1. **Low K regime** ($K \leq 5$): At small K, the platform factor would otherwise dominate, leaving too few factors for biology. The cohort indicator allows the model to reach $K_{\text{eff}} > 0$ under short iteration budgets.
2. **Training cohorts differ in platform** (RNA-seq vs microarray vs proteomics): The benefit is greatest when the offset is a simple mean shift (RNA-seq vs microarray, both measuring transcript abundance).
3. **Short iteration budgets** (quick fits, CV folds): The cohort indicator helps the CAVI escape the genomics-only local minimum faster.

Do not expect the cohort extension to resolve $\beta \to 0$ in the YFB model on merged TCGA+CPTAC data. The YFB fix should come from better β initialization (Solution 1 above) or preprocessing alignment (Solution 2).

8.2 Next Steps

1. **Implement Cox warm-start for YFB** (`cox_warmstart=TRUE` using SVD-initialized F). This is the most promising near-term fix for YFB $\beta \to 0$.
2. **Test cohort extension at $K = 5$** (run `run_cohort_lmm_benchmark.R` with `K_LB=5`, full convergence). The low-K benefit hypothesis is testable immediately.
3. **Consider the preprocessing-first approach**: z-standardize within each cohort *before* fitting, then use `cohort_id` to capture residual between-platform effects.

---

9 Appendix: Implementation Details

9.1 Corner-Point Encoding

```
# Cohort design matrix (C-1 columns; reference = first alphabetical level)
L_cohort <- model.matrix(~ factor(cohort_id))[, -1, drop = FALSE]
```

With $C = 2$ cohorts (TCGA, CPTAC), this produces a single binary column: 1 for CPTAC, 0 for TCGA (reference). With $C = 3$, two binary columns; and so on.

9.2 Normal Conjugate Update for F_cohort

For cohort column c and gene j, given residual $R_j^{(-c)} = Y_j - LF_j^\top - \sum_{c' \neq c} \ell_{c'} f_{c',j}$:

$$A_{cj} = \sigma_{F,c}^{-2} + n_c \hat{\tau}_j, \qquad B_{cj} = \hat{\tau}_j \cdot \ell_c^\top R_j^{(-c)}$$
$$\mathbb{E}_q[f_{c,j}] = B_{cj}/A_{cj}, \qquad \mathbb{E}_q[f_{c,j}^2] = (\mathbb{E}_q[f_{c,j}])^2 + 1/A_{cj}$$

9.3 KL Contribution to ELBO

$$-\mathrm{KL}[\mathcal{N}(\mu_{cj}, A_{cj}^{-1}) \,\|\, \mathcal{N}(0, \sigma_{F,c}^2)] = \frac{1}{2}\left(1 - \frac{\mathbb{E}[f_{cj}^2]}{\sigma_{F,c}^2} - \log(A_{cj}\sigma_{F,c}^2)\right)$$

This quantity is negative ($\mathrm{KL} \geq 0$) and is **added** to `elbo_full` by the caller — same sign convention as the EBNM KL terms.

9.4 ZF Restriction (YFB Model)

In `fit_cox_on_yf.R`, ZF is computed from the K biological columns only:

```
# EF_aug = cbind(EF, EF_cohort)  [p × (K + C_cols)]
# ZF must use only columns 1:K (biological factors)
ZF <- Y %*% EF_aug[, seq_len(K), drop = FALSE]  # n × K
```

This restriction applies in two locations: the inner CAVI loop (per-factor update) and Phase C (training concordance sign check). If `C_cols > 0` and the restriction is omitted, `EF_norms` has length $K + C_{\text{cols}}$, contaminating the β update (test YFBCohort-T4 catches this). “‘